

Shamsuddeen Hassan Muhammad, PhD

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Education

- 2023  **PhD in Computer Science, University of Porto, Portugal.**
Thesis Title: *Domain-Specific and Context-Aware Approaches to Sentiment Analysis.*
Main Supervisor: [Pavel Brazdil](#)
Co-Supervisor: [Alípio Jorge](#)
- 2013  **MSc in Computer Science, University of Manchester, UK.**
Dissertation Title: *Error-Correcting Codes in Wireless Communication.*
Supervisor: [Barry Cheetham](#)
- 2009  **BSc in Computer Science, Bayero University, Kano.**
Best Student Award: Received award as second-best student.

Academic and Research Experience

- 2025 – ...  **Visiting Lecturer (Natural Language Processing & Large Language Models)**
African Institute for Mathematical Sciences (AIMS), South Africa
-  **Visiting Lecturer (Natural Language Processing & Large Language Models)**
African Institute for Mathematical Sciences (AIMS), Cameroon
- 2024 – ...  **Advanced Research Fellow/Google DeepMind Academic Fellow**
Department of Computer Science
Imperial College London, UK
-  **Visiting Lecturer (Natural Language Processing)**
York St John University, UK
- 2023 – ...  **Senior Lecturer**
Department of Computer Science, Bayero University, Kano, Nigeria
- 2017 – 2023  **Lecturer I**
Department of Computer Science, Bayero University, Kano, Nigeria
- 2014 – 2017  **Assistant Lecturer**
Department of Computer Science, Bayero University, Kano, Nigeria
- 2012 – 2014  **Graduate Assistant**
Department of Computer Science, Bayero University, Kano, Nigeria
- 2020 – 2022  **NLP Researcher**
INESC TEC, Porto, Portugal

Research Interest

- **Computational Social Science:** Sentiment Analysis, Hate Speech Detection, Emotion Analysis
- **Low-Resource Languages:** Dataset Creation, Data Annotation, LLM Evaluation, and Adaptation
- **Multilingual NLP:** Enhancing Models for Underrepresented Languages and Cultures

Grants, Fellowships & Honors

Research Grants

- 2025
- **Co-PI:** Generative AI for Anxiety, Depression and Psychosis — Wellcome Trust. Developing and evaluating generative AI models to safely improve the measurement or treatment of mental health conditions, working in collaboration with clinicians and people with lived experience. (Up to £2,700,000)
 - **Co-PI:** Cross-Cultural and Multilingual Mental Health Safety — OpenAI. Researching and building safeguards for AI systems to ensure they are safe and effective for mental health applications across diverse languages and cultural contexts. (\$USD 50,000)
 - **Principal Investigator (PI):** Open Data Collection Playbook & Collection Platform — Developing an open-source modular data collection playbook and platform to enable grassroots NLP data collection in low-resource language communities (\$USD 430,000)
- 2023
- **Principal Investigator (PI):** Nigerian AI Research Grant — Automatic Offensive Language Detection for Nigerian Languages (₦5,000,000)
 - **Co-PI:** Google Research *exploreCSR* Award (USD 18,000)
 - **Co-PI:** Google Research *TensorFlow* Award (USD 10,000)
- 2022
- **PI:** Artificial Intelligence for Education Innovation Research Network in Africa (40,000 USD)
 - **PI:** Lacuna Fund 2021 Language Datasets Award — Development of Hate Speech Dataset for African Languages (147,000 USD)
 - **PI:** Oracle Cloud Infrastructure (OCI) Award (50,000 USD)
 - **Co-Investigator (Co-I):** Lacuna Fund — Developing Nigerian Speech Corpus (120,000 USD)
 - **Co-PI:** Google Research *exploreCSR* Award (18,000 USD)
 - **Co-PI:** *TensorFlow* Award (10,000 USD)
- 2021
- **PI:** Lacuna Fund 2020 Language Datasets Award — Development of Sentiment Analysis Dataset for African Languages (60,000 USD)
 - **PI:** Lacuna Fund — SemEval Shared Task Support (8,500 USD) —

Awards

- 2025
- **ACL 2025 Best Resource Paper Award** — BRIGHTER: Textual Emotion Recognition Datasets for 28 Languages.

Awards (continued)

- **SemEval 2025 Best Task Paper Award** — Task 11: Bridging the Gap in Text-Based Emotion Detection.
- **NAACL 2025 Outstanding Paper Award** — IrokoBench: A New Benchmark for African Languages in the Age of LLMs.
- **Deep Learning Indaba 2025 Best Paper Award** — The State of Large Language Models for African Languages.
- 2024 ■ **NeurIPS 2024 (C3NLP Workshop) Best Non-archival Paper Award** — BLEnD: Everyday Knowledge in Diverse Cultures and Languages.
- **SemEval 2024 Honourable Mention** — *Best Task Paper* (Semantic Textual Relatedness Task).
- 2023 ■ **AfricaNLP 2023 (Kigali) Best Non-archival Paper Award** — AfriSenti: A Twitter Sentiment Analysis Benchmark for African Languages.
- **IJCNLP-AACL 2023 Area Chair Award** — MasakhaNER: Named Entity Recognition for African Languages.
- 2020 ■ **EMNLP 2020 (Findings) Wikimedia Research Award** — Participatory Research for Low-resourced Machine Translation.

Research Impact

H-index: 22 **Citations:** 5,055 (as of May 19, 2026)

Teaching Experience

- Large Language Model: Visiting Lecturer, AIMS Cameroon (2026)
- Natural Language Processing: University of York St John (2026)
- Natural Language Processing: University of York St John (2026)
- Natural Language Processing: Bayero University, Kano (2025)
- Natural Language Processing: Visiting Lecturer, AI for Science, Africa Institute of Mathematical Sciences (AIMS), South Africa (2025)
- Natural Language Processing: Visiting Lecturer, AIMS Cameroon (2025)
- Artificial Intelligence Concept: Lecturer, York St John University, London Campus (2024)
- Machine Learning: Lecturer, York St John University, London Campus (2024)
- Machine Learning: Teaching Assistant, Imperial College London (2024)
- Deep Learning: Lead Teaching Assistant, NeuroMatch Academy (2023)
- Python Programming: Lecturer, Bayero University, Kano (2017)
- Data Structures and Algorithms: Lecturer, Bayero University, Kano (2016)
- Data Mining: Lecturer, Bayero University, Kano (2016)

Selected Student Supervision

PhD Students

- 2024 – Present
- **Sukhairja Muhammd** — *Bayero University, Kano*
Topic: “Automatic Speech Recognition for African languages”
 - **Murja Gadanya** — *Bayero University, Kano*
Topic: “Multi-modal emotion recognition in low-resource African languages”
 - **Muhammad Yobe** — *Bayero University, Kano*
Topic: “Automatic Detection of Machine-Generated Content in Low-Resource Languages”

MSc Students

- 2025 – 2026
- **Stefan Creasta** — *Imperial College London*
Topic: “Error Detection and Correction using LLMs”
 - **Paula Anyanwu** — *African Institute of Mathematical Science (AIMS) Camerrou*
Topic: “Red-Teaming and Safety Evaluation of LLMs in Multilingual Settings”
 - **Valdini Douglass Lemofouet** — *African Institute of Mathematical Science (AIMS) Camerrou*
Topic: “Aligning Large Language Models for Safety in African Languages?” — *African Institute of Mathematical Science (AIMS) Camerrou*
 - **Ngozi Uzor** — *African Institute of Mathematical Science (AIMS) Camerrou*
Topic: “Evaluating LLMs for low-resource African languages”

Teaching Certificates

- 2019
- **Teaching in Higher Education, University of Porto, Portugal.**
Certificate for teaching in higher education.
- 2020
- **Certified Instructor, The Carpentries.**
Certified instructor in teaching foundational computational and data skills to researchers using R and Python programming languages.

Academic Service, Leadership & Community

University Service

- Imperial College London — Grant Review Committee Member
- Bayero University, Kano — Senate Member
- Bayero University, Kano — Faculty Examination Lead
- Bayero University, Kano — BSc Computer Science with Economics, Course Leader

Scholarly Community Service

- Area Chair — ACL/EMNLP/NAACL
- Program Committee / Reviewer — ACL, EMNLP, NAACL, COLING, LREC, AfricaNLP

Academic Service, Leadership & Community (continued)

Leadership, Community & Organization

- 2022–...  Founder — [Arewa Data Science](https://arewadatascience.github.io) (arewadatascience.github.io)
A community empowering underrepresented students to build careers in data science and AI through free training and mentorship.
-  Founder — [HausaNLP](#)
An open-source initiative advancing NLP for Hausa, one of the most widely spoken languages in Africa.
-  Organizer / Chair — [AfricaNLP Workshop](#), the largest NLP venue for African languages
-  Shared Task Organizer — SemEval (2023–2026): [AfriSenti](#), [SemRel](#), [Emotion Recognition](#), [DimABSA](#), and [POLAR](#), advancing NLP evaluation for low-resource African languages
-  Shared Task Organizer — SemEval Shared Tasks (2023, 2024, 2025, 2026)
- 2022–...  Active Member — MasakhaneNLP
-  Active Member — Black in AI

List of Publications

Shamsuddeen Hassan Muhammad
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November 5, 2025

Awarded Publications

1. **Muhammad, S. H.**, Ousidhoum, N., Abdulmumin, I., *et al.* (2025). BRIGHTER: Bridging the Gap in Human-Annotated Textual Emotion Recognition Datasets for 28 Languages. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025)* (pp. 8895–8916). Vienna, Austria. [**ACL 2025 Best Resource Paper Award**]
2. **Muhammad, S. H.**, Ousidhoum, N., Abdulmumin, I., *et al.* (2025). SemEval-2025 Task 11: Bridging the Gap in Text-Based Emotion Detection. In *Proceedings of the 19th International Workshop on Semantic Evaluation (SemEval-2025)*. [**Best Task Paper Award**]
3. **Muhammad, S. H.**, Abdulmumin, I., Ayele, A. A., Ousidhoum, N., Adelani, D. I., *et al.* (2023). AfriSenti: A Twitter Sentiment Analysis Benchmark for African Languages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023, Main)* (pp. 13968–13981). Singapore. [**Best Non-archival Paper Award at AfricaNLP 2023 Workshop**]
4. Ousidhoum, N., **Muhammad, S. H.**, *et al.* (2024). SemEval Task 1: Semantic Textual Relatedness for African and Asian Languages. In *Proceedings of the 18th International Workshop on Semantic Evaluation (SemEval-2024)* (pp. 1963–1978). Mexico City, Mexico. Association for Computational Linguistics. [**Honourable Mention, Best Task Paper Award**]
5. Adelani, D. I., Ojo, J., Azime, I. A., Zhuang, J. Y., Alabi, J. O., He, X., Ochieng, M., Hooker, S., Bukula, A., Lee, E.-S. A., Chukwuneke, C. I., Buzaaba, H., Sibanda, B. K., Kalipe, G. K., Mukiibi, J., Kabongo Kabenamualu, S., Yuehgoh, F., Setaka, M., Ndolela, L., Odu, N., Mabuya, R., Osei, S., **Muhammad, S. H.**, Samb, S., Guge, T. K., Sherman, T. V., & Stenetorp, P. (2025). IrokoBench: A New Benchmark for African Languages in the Age of Large Language Models. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL 2025, Long Papers)* (pp. 2732–2757). Albuquerque, New Mexico. [**Outstanding Paper Award**]

6. Hussen, K. Y., Sewunetie, W. T., Ayele, A. A., Imam, S. H., **Muhammad, S. H.**, & Yimam, S. M. (2025). The State of Large Language Models for African Languages: Progress and Challenges. *arXiv preprint arXiv:2506.02280*. [**Best Paper Award at Deep Learning Indaba 2025**]
7. Myung, J., Lee, N., Zhou, Y., Jin, J., Putri, R. A., Antypas, D., Borkakoty, H., Kim, E., Perez-Almendros, C., Ayele, A. A., Gutiérrez-Basulto, V., Ibáñez-García, Y., Lee, H., **Muhammad, S. H.**, Park, K., Rzaev, A. S., White, N., Yimam, S. M., Pilehvar, M. T., Ousidhoum, N., Camacho-Collados, J., & Oh, A. (2024). BLEND: A Benchmark for LLMs on Everyday Knowledge in Diverse Cultures and Languages. In *Advances in Neural Information Processing Systems (NeurIPS 2024, Datasets and Benchmarks Track)* (pp. 78104–78146). Curran Associates, Inc. [**Best Non-archival Paper Award at C3NLP Workshop**]
8. Nekoto, W., Marivate, V., Matsila, T., Fasubaa, T., Fagbohunbe, T., Akinola, S. O., **Muhammad, S. H.**, *et al.* (2020). Participatory Research for Low-resourced Machine Translation: A Case Study in African Languages. In *Findings of the Association for Computational Linguistics: EMNLP 2020* (pp. 2144–2160). Online. Association for Computational Linguistics. [**Wikimedia Research Award**]
9. Adelani, D. I., Neubig, G., Ruder, S., Rijhwani, S., Beukman, M., **Muhammad, S. H.**, *et al.* (2022). MasakhaNER 2.0: Africa-centric Transfer Learning for Named Entity Recognition. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022)* (pp. 4488–4508). Abu Dhabi, United Arab Emirates. Association for Computational Linguistics. [**Service: IJCNLP–AAACL 2023 Area Chair Award — associated with MasakhaNER 2.0**]
10. **Muhammad, S. H.**, Abdulmumin, I., Ayele, A. A., Adelani, D. I., Ahmad, I. S., *et al.* (2025). AfriHate: A Multilingual Collection of Hate Speech and Abusive Language Datasets for African Languages. In *Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL 2025, Long Papers)* (pp. 1854–1871). Albuquerque, New Mexico. Association for Computational Linguistics. [**IRCAI Global Top 100 AI Projects Award for advancing the UN Sustainable Development Goals.**]

Other Publications

11. **Workshop, BigScience**, Scao, T. L., Fan, A., Akiki, C., Pavlick, E., Ilić, S., Hesslow, D., Castagné, R., Luccioni, A. S., Yvon, F., *et al.* (2022). BLOOM: A 176B-Parameter Open-Access Multilingual Language Model. In *arXiv preprint arXiv:2211.05100*.
12. **Muhammad, S. H.**, Ahmad, I. S., Abdulmumin, I., Lawan, F. I., Sani, B., Imam, S. H., Aliyu, Y., Sani, S. A., Umar, A. U., Gwadabe, T., *et al.* (2025). HausaNLP: Current Status, Challenges and Future Directions for Hausa Natural Language Processing. *arXiv preprint arXiv:2505.14311*.
13. Imam, S. H., Sani, B., Gete, D. K., Ahamed, B. Y., Ahmad, I. S., Abdulmumin, I., Yimam, S. M., Bello, M. Y., **Muhammad, S. H.**, *et al.* (2025). Automatic Speech Recognition for African Low-Resource Languages: Challenges and Future Directions. In *Proceedings of the AfricaNLP 2025 Workshop at ACL 2025*.

14. Sani, B., Soy, A., Imam, S. H., Mustapha, A., Aliyu, L. J., Abdulmumin, I., Ahmad, I. S., **Muhammad, S. H.**, *et al.* (2025). Who Wrote This? Identifying Machine vs Human-Generated Text in Hausa. In *Proceedings of the AfricaNLP 2025 Workshop at ACL 2025*.
15. Yu, H., Alabi, J. O., Bukula, A., Zhuang, J. Y., Lee, E.-S. A., Guge, T. K., Azime, I. A., Buzaaba, H., Sibanda, B. K., Kalipe, G. K., Mukiibi, J., Kabongo Kabenamualu, S., Setaka, M., Ndoela, L., Odu, N., Mabuya, R., **Muhammad, S. H.**, *et al.* (2025). INJONGO: A Multicultural Intent Detection and Slot-filling Dataset for 16 African Languages. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025, Long Papers)* (pp. 9429–9452). Vienna, Austria. Association for Computational Linguistics.
16. Naseem, U., Ren, J., Anwar, S., Kohail, S., Veliz, R. A. G., Geislinger, R., Jabr, A., Abdulmumin, I., Qureshi, L., Borkar, A. A., **Muhammad, S. H.**, *et al.* (2025). POLAR: A Benchmark for Multilingual, Multicultural, and Multi-Event Online Polarization. *arXiv preprint arXiv:2505.20624*.
17. Belay, T. D., Azime, I. A., Ahmad, I. S., Adelani, D. I., Abdulmumin, I., Ayele, A. A., **Muhammad, S. H.**, & Yimam, S. M. (2025). AfroXLMR-Social: Adapting Pre-trained Language Models for African Languages Social Media Text. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025)*. Association for Computational Linguistics.
18. Alabi, J. O., Azime, I. A., Zhang, M., España-Bonet, C., Bawden, R., Zhu, D., Adelani, D. I., Odoje, C. O., Akinade, I., Maab, I., **Muhammad, S. H.**, *et al.* (2025). AfriDoc-MT: Document-level MT Corpus for African Languages. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025 (main))*. Association for Computational Linguistics.
19. Ousidhoum, N., **Muhammad, S. H.**, *et al.* (2024). SemRel2024: A Collection of Semantic Textual Relatedness Datasets for 13 Languages. In *Findings of the Association for Computational Linguistics: ACL 2024* (pp. 2512–2530). Bangkok, Thailand. Association for Computational Linguistics.
20. **Muhammad, S. H.**, Adelani, D. I., Ruder, S., Ahmad, I. S., Abdulmumin, I., *et al.* (2022). NaijaSenti: A Nigerian Twitter Sentiment Corpus for Multilingual Sentiment Analysis. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC 2022)* (pp. 590–602). Marseille, France. European Language Resources Association.
21. Bayes, E., Azime, I. A., Alabi, J. O., Kgomo, J., Eloundou, T., Proehl, E., Chen, K., Khadir, I., Etori, N. A., **Muhammad, S. H.**, *et al.* (2024). Uhura: A Benchmark for Evaluating Scientific Question Answering and Truthfulness in Low-resource African Languages. *arXiv preprint arXiv:2412.00948*.
22. Wang, J., Adelani, D. I., Agrawal, S., Masiak, M., Rei, R., Briakou, E., Carpuat, M., He, X., Bourhim, S., Bukula, A., Mohamed, M., Olatoye, T., Adewumi, T., Mokayed, H., Mwase, C., Kimotho, W., Yuehgoh, F., Aremu, A., Ojo, J., **Muhammad, S. H.**, *et al.* (2024). AfriMTE and AfriCOMET: Enhancing COMET to Embrace Under-resourced African Languages. In *Proceedings of the 2024 Conference of the North*

American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL 2024, Long Papers) (pp. 5997–6023). Mexico City, Mexico. Association for Computational Linguistics.

23. Abdulmumin, I., Mkhwanazi, S., Mbooi, M., **Muhammad, S. H.**, Ahmad, I. S., *et al.* (2024). Correcting FLORES Evaluation Dataset for Four African Languages. In *Proceedings of the Ninth Conference on Machine Translation (WMT 2024)* (pp. 570–578). Miami, Florida, USA. Association for Computational Linguistics.
24. Parida, S., Abdulmumin, I., **Muhammad, S. H.**, Bose, A., Kohli, G. S., Ahmad, I. S., *et al.* (2023). HaVQA: A Dataset for Visual Question Answering and Multi-modal Research in Hausa Language. In *Findings of the Association for Computational Linguistics: ACL 2023* (pp. 10162–10183). Toronto, Canada. Association for Computational Linguistics.
25. Vargas, F. A., Guimarães, S., **Muhammad, S. H.**, Alves, D., Ahmad, I. S., *et al.* (2024). HausaHate: An Expert Annotated Corpus for Hausa Hate Speech Detection. In *Proceedings of the 8th Workshop on Online Abuse and Harms (WOAH 2024)*.
26. Parida, S., Bojar, O., Abdulmumin, I., **Muhammad, S. H.**, Ahmad, I. S., *et al.* (2024). Findings of WMT2024 English-to-Low Resource Multimodal Translation Task. In *Proceedings of the Ninth Conference on Machine Translation (WMT 2024)*.
27. Wang, J., Adelani, D. I., Agrawal, S., Rei, R., Briakou, E., **Muhammad, S. H.**, *et al.* (2023). AfriMTE and AfriCOMET: Empowering COMET to Embrace Under-resourced African Languages. In *Proceedings of the 2023 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2023)*.
28. Adewumi, T. P., Adeyemi, M., Anuoluwapo, A., Peters, B., Buzaaba, H., **Muhammad, S. H.**, *et al.* (2023). AfriWOZ: Corpus for Exploiting Cross-Lingual Transfer for Dialogue Generation in Low-Resource African Languages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*.
29. **Muhammad, S. H.**, Brazdil, P., & Jorge, A. M. (2023). Combining Symbolic and Deep Learning Approaches for Sentiment Analysis. In *Compendium of Neurosymbolic Artificial Intelligence*.
30. Dione, C. M. B., Adelani, D. I., Nabende, P., Alabi, J. O., Sindane, T., **Muhammad, S. H.**, *et al.* (2023). MasakhaPOS: Part-of-Speech Tagging for Typologically Diverse African Languages. In *Proceedings of the 2023 Annual Meeting of the Association for Computational Linguistics (ACL 2023)*.
31. Nekoto, W. O., Marivate, V., Matsila, T., Fasubaa, T. E., Kolawole, T., **Muhammad, S. H.**, *et al.* (2020). Participatory Research for Low-resourced Machine Translation: A Case Study in African Languages. *arXiv preprint arXiv:2010.02353*.
32. **Muhammad, S. H.** (2019). An Overview of Sentiment Analysis Approaches. *arXiv preprint*.
33. **Muhammad, S. H.**, Abdulmumin, I., Ahmad, I. S., Gwadabe, T. R., *et al.* (2023). AfriSenti: A Twitter Sentiment Analysis Benchmark for African Languages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*.

34. Aliyu, S. M., Abdulmumin, I., **Muhammad, S. H.**, Ahmad, I. S., Salahudeen, S. A., *et al.* (2023). HausaNLP at SemEval-2023 Task 10: Transfer Learning, Synthetic Data and Side-information for Multi-level Sexism Classification. *arXiv preprint*.
35. Ousidhoum, N., **Muhammad, S. H.**, Abdalla, M., Abdulmumin, I., Ahmad, I. S., *et al.* (2023). AfriQA: Cross-lingual Open-Retrieval Question Answering for African Languages. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP 2023)*.
36. Emezue, C. C., Nigatu, H. H., Thinwa, C., Zhou, H., **Muhammad, S. H.**, *et al.* (2023). The African Stopwords Project: Curating Stopwords for African Languages. *arXiv preprint*.
37. Abdulmumin, I., Beukman, M., Alabi, J. O., Emezue, C. C., **Muhammad, S. H.**, *et al.* (2022). Separating Grains from the Chaff: Using Data Filtering to Improve Multilingual Translation for Low-Resourced African Languages. *arXiv preprint*.
38. Adelani, D. I., Neubig, G., Ruder, S., Rijhwani, S., Beukman, M., **Muhammad, S. H.**, *et al.* (2022). MasakhaNER 2.0: Africa-centric Transfer Learning for Named Entity Recognition. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022)*.
39. Meyer, J., Adelani, D. I., Casanova, E., Oktem, A., **Muhammad, S. H.**, *et al.* (2022). BibleTTS: A Large, High-Fidelity, Multilingual, and Uniquely African Speech Corpus. *arXiv preprint*.
40. **Muhammad, S. H.**, Brazdil, P., & Jorge, A. M. (2022). Symbolic Versus Deep Learning Techniques for Explainable Sentiment Analysis. In *Portuguese Conference on Artificial Intelligence*.
41. Abdulmumin, I., Dash, S. R., Dawud, M. A., Parida, S., **Muhammad, S. H.**, *et al.* (2022). Hausa Visual Genome: A Dataset for Multi-Modal English to Hausa Machine Translation. In *Proceedings of the 2022 International Conference on Language Resources and Evaluation (LREC 2022)*.
42. Aliyu, S. M., Wajiga, G. M., Murtala, M. T., **Muhammad, S. H.**, *et al.* (2022). HERDPhobia: A Dataset for Hate Speech Against Fulani in Nigeria. *arXiv preprint*.
43. **Muhammad, S. H.**, Brazdil, P., & Jorge, A. M. (2020). Incremental Approach for Automatic Generation of Domain-Specific Sentiment Lexicon. In *Advances in Information Retrieval (ECIR 2020)* (pp. 619–623).
44. Ahmad, I. S., Abu Bakar, A., & Yaakub, M. R., **Muhammad, S. H.** (2020). A Survey on Machine Learning Techniques in Movie Revenue Prediction. *SN Computer Science*, 1.
45. Caswell, I., Kreutzer, J., Wang, L., Wahab, A., van Esch, D., **Muhammad, S. H.**, *et al.* (2021). Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets. *Transactions of the Association for Computational Linguistics*, 10, 50–72.
46. **Muhammad, S. H.**, Mustapha, A., & Haruna, K. (2017). Massive Open Online Courses: Awareness, Adoption, Benefits and Challenges in Sub-Saharan Africa. In *Proceedings of OcRI*.

47. Mustapha, A., **Muhammad, S. H.**, & Salahudeen, S. A. (2016). Massive Open Online Courses: A Success of Cloud Computing in Education. In *Proceedings of OcRI*.
48. Haruna, K., Baffa, M., **Muhammad, S. H.**, & Abubakar, U. B. (2017). A Framework for Implementation of E-Classroom System. In *Proceedings of OcRI*.